

CA 7101 User Interface Design



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INSTRUCTOR

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(Please Email/Facebook me first to make sure that I will be there.)

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Course Syllabus and Lectures

All course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/CA7101/CS-CA7101.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/CA7101/L1.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CA7101/L2.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CA7101/L3.pdf>
 - etc

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Textbooks

- Reality+ : Virtual Worlds and the Problems of Philosophy, David J. Chalmers, London, UK (2022)
- Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, 2nd Edition (2016)
- สื่อดิจิทัลใหม่...สื่อแห่งอนาคต, ชุตินันต์ เกิดวิบูลย์เวช คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ (นิด้า) 2560

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Related-Academic Papers

- Chutisant Kerdvibulvech: Location-Based Augmented Reality Games Through Immersive Experiences. HCI (15) 2021: 452-461.
- N. Kaewpanukrangsri and C. Kerdvibulvech, "Gesture-Based Communication via User Experience Design: Integrating Experience into Daily Life for an Arm-Swing Input Device," Augmented Human Research, Vol. 4, Issue 2, Springer Publisher (Ingenta Connect, EBSCO Discovery Service, ProQuest-ExLibris Primo, ProQuest-ExLibris Summon, OCLC WorldCat Discovery Service, and Google Scholar), April 2019.
- Chutisant Kerdvibulvech, Sheng-Uei Guan: Affective Computing for Enhancing Affective Touch-Based Communication Through Extended Reality. ICCSA (2) 2019: 351-360
- C. Kerdvibulvech, "A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models," EURASIP Journal on Embedded Systems (JES), 9 pages, No. 18, Springer Publisher (Elsevier's Scopus, EiCompendex, INSPEC, SciFinder, SCImago, ACM, DBLP, and DOAJ), 2014.
- เอกสารคำสอน นน 7005 การผลิตสื่อดิจิทัล คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ โดย รศ.ดร.ชุตินันต์ เกติวิบูลย์เวช.

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Websites and Links

- <https://www.sciencedirect.com/>
- <https://www.scimagojr.com/>
- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/>

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Overview

- Class Projects
- Introduction to Technologies for Communication

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Class Overview

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Evaluation

- Group Research Project + Individual Project 85 %
- Applied Rinkou 5 %
- Dream Jobs 5 %
- Class Attendance 5 %

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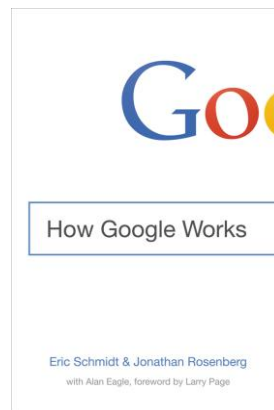
Projects

- Introduction (About 1-2 weeks)
- Group Research Project (About 8-10 weeks)
 - Touch Communication (National University of Singapore)
 - UX-UI / Metaverse / Artificial Intelligence / Deep Learning / etc
- Individual Project (About 3-4 weeks)
 - Website, UX-UI, and SEO (Search Engine Optimization)
- Field Trip(s) (About 1-2 weeks)

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Applied Rinkou

- <https://www.amazon.com/How-Google-Works-Eric-Schmidt/dp/1455582344>
- 8 Chapters
- 20-25 mins/Week



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Dream Jobs

- Big Five or GAMAM defines Google, Amazon, Meta, Apple, and Microsoft as the tech giants
- In each group, select three recent dream jobs about *Reality Technologies and Rapid User Experience Design* or related technologies from Big Five within the past three years
- Presentation
 - Explaining job descriptions, essential employability skills, and self-assessment for 20 mins
 - Pitching yourselves to interviewers for 3 mins

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Dream Jobs (cont.)

- Group discussion to select three recent dream jobs from Big Five within the past three years
- ...JobPosition1..., ...Company1...
...JobPosition2..., ...Company2...
...JobPosition3..., ...Company3...
- Different job positions and companies in each group

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Technologies for Communication

Digital media are any media that are encoded in a machine-readable format. Digital media can be created, viewed, distributed, modified and preserved on digital electronics devices.

["Digital Media"](#) (PDF). *Technology Brief*. University of Guelph. September 2006. Retrieved 28 March 2014.

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



aurora.dawn.com

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Future of Digital Media?



www.3dsg.com.au

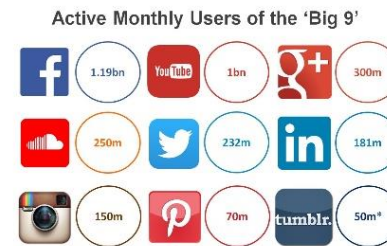
- Big Data / SEO (Search Engine Optimization) / SEM (Search Engine Marketing)
- AI (Artificial Intelligence) / Deep Learning
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality) / VR (Virtual Reality)
 - New Digital Media

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Digital Media Trends for 2014

Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)
*Google+ data not reliable due to monthly user count and this figure is based on an estimate from
Facebook's data on 11 December 2013. But the source is data on 11 December 2013. KED's data per month.

kamber

::Source::

Kamber (Sydney, Australia)

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Digital Media Trends for 2016



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<https://www.facebook.com/zuck>

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Digital Media Trends for 2017/2018



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The Future of Digital Media

- Big Data / Cloud Computing / SEO (Search Engine Optimization) – SEM (Search Engine Marketing)
- From Mobile-Oriented to AI-Oriented
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)



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www.3dsg.com.au

Augmented Reality vs. Virtual Reality

Augmented Reality

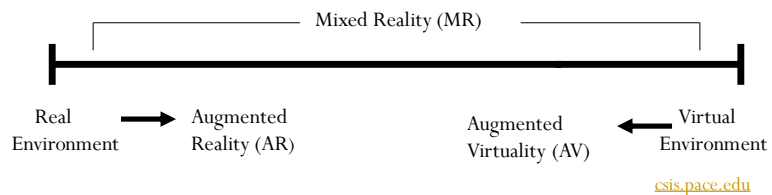
- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

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Miligram's Reality-Virtuality Continuum



Milgram coined the term “Augmented Virtuality” to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

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From Virtual Reality to Facebook360



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Virtual Reality



“The Oculus community is leading us into the future, and we can’t wait for everyone to experience the magic of VR and the next platform.”...**Mark Zuckerberg, CEO of Facebook**

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24 September 2015

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Chatbot

- A computer program which conducts a conversation via auditory or textual methods
- A chatter robot is a type of conversational agent, a computer program designed to simulate an intelligent conversation with one or more human users via auditory or textual methods
- AI (Artificial Intelligence) / Deep Learning

<https://en.wikipedia.org/wiki/Chatterbot>

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When Facebook understands our messages automatically..

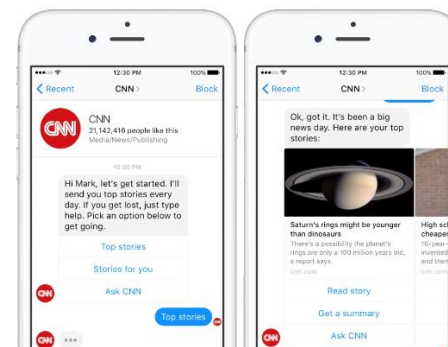
เมื่อเฟซบุ๊กเข้าใจข้อความของเราได้เอง (1) (2) (3)..หนังสือพิมพ์เดลินิวส์

- <http://www.dailynews.co.th/it/502607>
- <http://repository.nida.ac.th/bitstream/handle/662723737/3148/59104.pdf?sequence=1&isAllowed=y>
- <http://repository.nida.ac.th/handle/662723737/3151>

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Chatbot by CNN



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Thank you

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